

$$\text{typ}((R, x, v_r)) \triangleq R \quad \text{typ}((W, x, v_w)) \triangleq W \quad \text{typ}((U, x, v_r, v_w)) \triangleq U$$

$$\text{val}_r((R, x, v_r)) \triangleq v_r \quad \text{val}_r((U, x, v_r, v_w)) \triangleq v_r$$

$$\text{val}_w((W, x, v_w)) \triangleq v_w \quad \text{val}_w((U, x, v_r, v_w)) \triangleq v_w$$

$$\text{loc}((R, x, v_r)) = \text{loc}((W, x, v_w)) = \text{loc}((U, x, v_r, v_w)) \triangleq x$$

$$[A] \text{ for the identity relation on } A: [A] \triangleq \{(a, a) \mid a \in A\}$$

$r; r'$ for composition

$r^?$ for the reflexive closure of r

$$\text{irreflexive}(r) \stackrel{\text{def}}{\iff} \neg \exists a. (a, a) \in r$$

$$\text{acyclic}(r) \stackrel{\text{def}}{\iff} \text{irreflexive}(r^+)$$